Maxim Vochten

Senior Researcher | Research leadership | Robotics and autonomous systems

maxim.vochten@mil.be | maximvochten.github.io | Brussels, Belgium

PROFILE

Mechanical engineer and researcher who leads collaborative projects and supervises researchers while contributing to scientific work. Experience spans robot learning, manipulation and autonomy, from industrial applications to systems used in hazardous environments.

EXPERIENCE

Royal Military Academy, Brussels | Senior Researcher | Jan 2025 - present

- Lead RMA's team in GENIUS, an EDF project on robotic demining, with responsibility for technical work and collaboration with consortium partners.
- Serve as promotor alongside colleagues on ATTENTION (DEFRA) and eUGS (EDA), supporting research direction, researcher supervision and project delivery.
- Supervise researchers across robotics and autonomous systems activities and contribute to collaborative research proposals and publications.

KU Leuven | Postdoctoral Researcher | Oct 2018 - Dec 2024

- Developed methods for representing and generalising skills from human demonstrations in Prof. Joris De Schutter's ERC Advanced Grant ROBOTGENSKILL.
- Led the four-person KU Leuven team in PROROB, a Flanders Make ICON project on virtual-reality approaches to industrial robot programming.
- Co-promotor of two PhD researchers; supervised master's theses; coordinated teaching-assistant duties in the Automation division.

KU Leuven | Doctoral Researcher | Aug 2013 - Sep 2018

- Studied invariant representations of rigid-body motion for motion recognition and robot learning from demonstration; assisted with engineering teaching.

EDUCATION

PhD in Mechanical Engineering, KU Leuven (2018) | MSc in Mechanical Engineering, KU Leuven (2013)

SUPERVISION AND TEACHING

Co-promotor of two doctoral researchers and mentor or co-supervisor of more than 15 master's thesis students at KU Leuven. Teaching assistant for Applied Mechanics 3 and Mechanisms and Vibration; organiser of student robotics workshops.

SELECTED PUBLICATIONS

T.-T. Nguyen, **M. Vochten** et al., Predictive and Inertia-Aware Motion Planning for USV Navigation in Cluttered Harbors. IFAC World Congress, 2026.

T. T. Nguyen, **M. Vochten** et al., Deep Long-Term Ship Motion Prediction with Anomaly Detection for Autonomous Maritime UAV Landing. ISRM, 2026.

M. Vochten et al., Invariant Descriptors of Motion and Force Trajectories for Interpreting Object Manipulation Tasks in Contact. IEEE Transactions on Robotics, 2023.

M. Vochten et al., Shape-preserving and reactive adaptation of robot end-effector trajectories. IEEE Robotics and Automation Letters, 2021.

Fuller records: Google Scholar and RMA research profile.

RESEARCH TOOLS AND RECOGNITION

Open-source packages: invariants_py and invariants_mat. Best paper, Human-Friendly Robotics (2021); KUKA Innovation Award finalist (2016).